

Lab Activity 3.3: Construct form validation using Bootstrap and JavaScript

STUDENT INFORMATION

Name	Registration No	Date
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Learning Outcomes:

- a) Construct client-side form validation using JavaScript.
 - b) Construct Bootstrap form components for improved user experience.
 - c) Perform form submission and validation errors gracefully.
-

Step 1: Setting Up the Document Structure & Styling

Create a new file named `passport_form.html`. We will start by defining the **Purple Theme** using CSS variables and styling the passport photo container to meet the **35mm x 45mm** requirement.

HTML

```
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1.0">
  <title>Lab 3.3: Passport Application Form</title>
  <link
href="https://cdn.jsdelivrivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap
.min.css" rel="stylesheet">
<style>
  :root { --primary-purple: #6f42c1; --hover-purple: #59359a; --
light-purple: #f3e5f5; }
  body { background-color: var(--light-purple); padding: 40px 20px;
}
  .form-container { max-width: 800px; margin: auto; background:
white; padding: 40px; border-radius: 15px; border-top: 8px solid
var(--primary-purple); }
  .text-purple { color: var(--primary-purple); }
  .bg-purple { background-color: var(--primary-purple) !important;
}
  .btn-purple { background-color: var(--primary-purple); color:
white; }
```

```

        .btn-purple:hover { background-color: var(--hover-purple); color:
white; }
        .passport-photo { width: 3.5cm; height: 4.5cm; object-fit: cover;
border: 3px solid #f8f9fa; outline: 1px solid #dee2e6; display:
block; margin: 0 auto 10px auto; }
        .form-control:focus, .form-select:focus { border-color: var(--
primary-purple); box-shadow: 0 0 0 0.25rem rgba(111, 66, 193, 0.25);
}
        #rangeValue { background-color: var(--primary-purple); color:
white; padding: 2px 10px; border-radius: 10px; font-size: 0.9rem; }
        #successMessage { display: none; }
</style>
</head>

```

Step 2: Constructing the UI (Form Components)

Add the HTML body. This includes the **Success Alert**, **Passport Image**, **Floating Labels**, and **Input Groups**. Note the use of `novalidate` in the form tag to allow custom JavaScript handling.

HTML

```

<div class="container">
  <div class="form-container shadow-lg">

    <div class="mb-3">
      <a href="LAB2_1.html" class="text-decoration-none text-
purple small fw-bold">&larr; Back to Profile</a>
    </div>

    <h2 class="mb-4 text-center fw-bold text-purple">Passport
Application</h2>
    <hr class="mb-5">

    <div id="successMessage" class="alert alert-success alert-
dismissible fade show" role="alert">
      <strong>Success!</strong> Your application has been
validated and submitted gracefully.
      <button type="button" class="btn-close" data-bs-
dismiss="alert" aria-label="Close"></button>
    </div>

    <form class="row g-4 needs-validation" id="passportForm"
novalidate>

      <div class="col-12 text-center">
        
        <p class="text-muted small">Standard Passport Size
(35mm x 45mm)</p>
      </div>

      <div class="col-md-6">
        <div class="form-floating">

```

```

        <input type="text" class="form-control"
id="firstName" placeholder="First Name" required>
        <label for="firstName">First Name</label>
        <div class="invalid-feedback">First name is
required.</div>
    </div>
</div>

<div class="col-md-6">
    <div class="form-floating">
        <input type="text" class="form-control"
id="lastName" placeholder="Last Name" required>
        <label for="lastName">Last Name</label>
        <div class="invalid-feedback">Last name is
required.</div>
    </div>
</div>

<div class="col-md-12">
    <label for="userEmail" class="form-label fw-
bold">Contact Email</label>
    <div class="input-group has-validation">
        <span class="input-group-text bg-purple text-
white">@</span>
        <input type="email" class="form-control"
id="userEmail" placeholder="e.g. name@mail.com" required>
        <div class="invalid-feedback">Please enter a
valid email address.</div>
    </div>
</div>

<div class="col-md-6">
    <label for="citizenStatus" class="form-label fw-
bold">Citizenship</label>
    <select class="form-select" id="citizenStatus"
required>
        <option selected disabled value="">Select
status...</option>
        <option>Citizen</option>
        <option>Permanent Resident</option>
    </select>
    <div class="invalid-feedback">Selection
required.</div>
</div>

<div class="col-md-6">
    <label for="yearsResidency" class="form-label fw-
bold">
        Residency Years: <span id="rangeValue">25</span>
    </label>
    <input type="range" class="form-range" min="1"
max="50" value="25" id="yearsResidency">
</div>

<div class="col-md-12">
    <label class="form-label d-block fw-
bold">Gender</label>
    <div class="form-check form-check-inline">

```

```

        <input class="form-check-input" type="radio"
name="gender" id="male" required>
        <label class="form-check-label"
for="male">Male</label>
    </div>
    <div class="form-check form-check-inline">
        <input class="form-check-input" type="radio"
name="gender" id="female" required>
        <label class="form-check-label"
for="female">Female</label>
    </div>
    <div class="invalid-feedback">Please select a
gender.</div>
</div>

    <div class="col-12">
        <div class="form-check">
            <input class="form-check-input" type="checkbox"
value="" id="invalidCheck" required>
            <label class="form-check-label" for="invalidCheck">
                I confirm that all information provided is
accurate.
            </label>
            <div class="invalid-feedback">You must agree before
submitting.</div>
        </div>
    </div>

    <div class="col-md-6">
        <div class="form-floating">
            <input type="password" class="form-control"
id="password" placeholder="Password" required>
            <label for="password">Password</label>
            <div class="invalid-feedback">Please enter a
password.</div>
        </div>
    </div>

    <div class="col-md-6">
        <div class="form-floating">
            <input type="password" class="form-control"
id="confirmPassword" placeholder="Confirm Password" required>
            <label for="confirmPassword">Confirm
Password</label>
            <div class="invalid-feedback"
id="confirmFeedback">Passwords do not match!</div>
        </div>
    </div>

    <div class="col-12 mt-5 d-flex gap-2">
        <button class="btn btn-purple w-75 py-3 fw-bold
shadow-sm" type="submit">SUBMIT APPLICATION</button>
        <button class="btn btn-outline-secondary w-25 py-3
fw-bold shadow-sm" type="reset" id="resetBtn">RESET</button>
    </div>

</form>
</div>

```

```
</div>
```

Step 3: JavaScript Logic (Validation & Interaction)

Implement the script to handle **Range Slider** updates, **Password Matching**, and **Graceful Error Handling**.

HTML

```
<script>
  (function () {
    'use strict'

    const form = document.getElementById('passportForm');
    const successDiv = document.getElementById('successMessage');
    const resetBtn = document.getElementById('resetBtn');
    const rangeInput = document.getElementById('yearsResidency');
    const rangeValue = document.getElementById('rangeValue');

    // Update range value display
    rangeInput.addEventListener('input', function() {
      rangeValue.textContent = this.value;
    });

    // Form Validation Logic
    form.addEventListener('submit', function (event) {
      if (!form.checkValidity()) {
        event.preventDefault();
        event.stopPropagation();
        successDiv.style.display = 'none';
      } else {
        event.preventDefault();
        successDiv.style.display = 'block';
        window.scrollTo({ top: 0, behavior: 'smooth' });
      }
      form.classList.add('was-validated');
    }, false);

    // Reset Logic
    resetBtn.addEventListener('click', function() {
      form.classList.remove('was-validated');
      successDiv.style.display = 'none';
      rangeValue.textContent = "25"; // Reset range display to
default
    });

  })();
  (function () {
    'use strict'

    const form = document.getElementById('passportForm');
    const password = document.getElementById('password');
    const confirmPassword =
document.getElementById('confirmPassword');

    // Fungsi untuk menyemak padanan kata laluan
```

```

function validatePassword() {
    if (password.value !== confirmPassword.value) {
        // Jika tidak sama, set ralat custom
        confirmPassword.setCustomValidity("Passwords do not
match");
    } else {
        // Jika sama, kosongkan ralat (sah)
        confirmPassword.setCustomValidity("");
    }
}

// Jalankan fungsi setiap kali pengguna menaip
password.addEventListener('change', validatePassword);
confirmPassword.addEventListener('keyup', validatePassword);

// Kekalkan logik submit anda yang sedia ada
form.addEventListener('submit', function (event) {
    // validatePassword dipanggil sekali lagi sebelum submit
untuk kepastian
    validatePassword();

    if (!form.checkValidity()) {
        event.preventDefault();
        event.stopPropagation();
    } else {
        event.preventDefault();
        document.getElementById('successMessage').style.display =
'block';
        window.scrollTo({ top: 0, behavior: 'smooth' });
    }
    form.classList.add('was-validated');
}, false);

// ... (Logik Range dan Reset anda yang lain)
})();
</script>

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.b
undle.min.js"></script>

```

Task Recap for Students:

1. **Step 1:** Copy CSS to define the look and the specific **3.5cm x 4.5cm** image size.
2. **Step 2:** Construct the form using Bootstrap's grid and specialized components.
3. **Step 3:** Apply the JavaScript to enable live slider updates and ensure the passwords match before submission.
4. **Observation:** Notice how the `was-validated` class triggers red/green borders and messages gracefully without refreshing the page.

STEP-BY-STEP VALIDATION:

Please provide a **screenshot** for every completed step. For each image, write a short **observation** explaining the logic behind the code you just implemented. This helps solidify your mastery of Bootstrap!

Progress Step	Screenshot of Output	Your Observation & Understanding
Step 1		
Step 2		
<i>Step 3</i>		

5. SCREENSHOT OUTPUT (COMPLETE):

Desktop View

The desktop view shows a web application with a blue header bar. On the left is a logo with 'JS' and 'JAVASCRIPT'. On the right are navigation links: 'LAB 1A', 'LAB 1B', 'LAB 1C', 'LAB 1D', and 'REPORT'. The main content area has a light purple background. It contains the text 'JAVASCRIPT LAB ACTIVITY' and 'PREPARED BY: NUR AINA BINTI MOHD ISAMUDDIN (10DIT24F1059)'. Centered is a white card titled 'Passport Application' with a 'Back to Profile' link. The card features a placeholder image for a passport photo, followed by input fields for 'First Name' and 'Last Name', a 'Contact Email' field, a 'Citizenship' dropdown, a 'Residency Years' slider, a 'Gender' section with 'Male' and 'Female' radio buttons, a confirmation checkbox, and 'Password' and 'Confirm Password' fields. At the bottom of the card are 'SUBMIT APPLICATION' and 'RESET' buttons. A footer at the very bottom reads '© 2024 WEB DESIGN TECHNOLOGY | PREPARED BY: NUR AINA BINTI MOHD ISAMUDDIN'.

Mobile View

The mobile view shows the same application on a smaller screen. The header bar is blue with the 'JS JAVASCRIPT' logo and a hamburger menu icon. The main content area has a light purple background with the text 'JAVASCRIPT LAB ACTIVITY' and 'PREPARED BY: NUR AINA BINTI MOHD ISAMUDDIN (10DIT24F1059)'. Centered is a white card titled 'Passport Application' with a 'Back to Profile' link. The card features a placeholder image for a passport photo, followed by input fields for 'First Name' and 'Last Name', a 'Contact Email' field, a 'Citizenship' dropdown, a 'Residency Years' slider, a 'Gender' section with 'Male' and 'Female' radio buttons, a confirmation checkbox, and 'Password' and 'Confirm Password' fields. At the bottom of the card are 'SUBMIT APPLICATION' and 'RESET' buttons. A footer at the very bottom reads '© 2024 WEB DESIGN TECHNOLOGY | PREPARED BY: NUR AINA BINTI MOHD ISAMUDDIN'.

First Name

Last Name

Contact Email

@ e.g. name@mail.com

Citizenship

Select status... ▾

Residency Years: 25

Gender

☐ Male ☐ Female

☐ I confirm that all information provided is accurate.

Password

Confirm Password

SUBMIT APPLICATION **RESET**

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MOHD ISAMUDDIN

6. LAB SUBMISSION TASK

Ensure you have completed all steps successfully!

1. **Compress** your folder: RegNo_Date.zip.
2. **Report:** Create a PDF containing all screenshots and explanations:
Report_RegNo_Date.pdf.

Happy Coding! Success is in your hands!